Soft Matter

PAPER

View Article Online
View Journal | View Issue

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

Check for updates

Cite this: *Soft Matter*, 2022, **18**, 2054

Column to pile transition in quasi-static deposition of granular chains†

Palash S. Sarate,[a] Tejas G. Murthy[b] and Prerna Sharma *[ac]

Received 28th October 2021,
Accepted 10th February 2022

DOI: 10.1039/d1sm01539g

rsc.li/soft-matter-journal

The repose angle is a key geometric property that characterises the inter-particle friction and thereby granular flows. One of the common methods to measure this property is to deposit a pile by extracting a pre-filled cylinder and letting the material flow out. While the repose angle of spherical beads is insensitive to the aspect ratio of this pre-filled column, we find that long flexible granular chains show a remarkable transition from stable vertical columns to conical piles depending on the aspect ratio. Below a critical aspect ratio, the cessation of flow of granular chains due to inter-chain entanglement stabilises the columns, while above the critical aspect ratio the conical piles of long granular chains arise not out of shear flow but instead through a series of column collapse instabilities during the deposition process. We also identify the critical chain length below which the granular chains flow and behave similar to spherical particles.

1 Introduction

Avalanches, segregation and stratification which are unique features of granular materials can be ascertained to some degree through an estimate of "geometric quantity" known as the repose angle.[1–7] The repose angle is the maximum slope angle at which a granular ensemble is at rest. This critical angle marks the boundary below which the material is in static mechanical equilibrium and above which it flows. The repose angle characterizes the inter-particle friction and naturally depends on the particle characteristics and morphology, interface friction, moisture content, roughness of the substrate on which the material reposes, *etc.*[8–17] Despite such multiple dependencies, repose is still used as one of the most important descriptors of the physico-mechanical properties of a granular ensemble. Several laboratory protocols have been developed to standardise the measurement of this angle depending on the property to be investigated.[18] The fundamental physics of pile formation is rooted in statistical mechanics of self-organized criticality, a theme which has been extensively studied theoretically but much less experimentally.[19–21]

The presence of moisture, chemical reactions, and mechanical forces often induces cohesion between particles in a granular ensemble. Such cohesive granular matter attain significantly higher mechanical rigidity than their dry, frictional counterparts due to attractive forces acting at the particle level.[10,22] Surprisingly, ensembles of dry and non-cohesive particles can also show similar mechanical rigidity to their cohesive counterpart when their shape is such that they can inter-penetrate and thus entangle. This is called "geometric cohesion" and has been observed in a variety of particle morphologies such as the U-shape, Z-shape, rod, chain, *etc.*[23–28] The effect of entanglement on the mechanical response of a system is well-understood for a drastically different length scale of molecular polymeric chains.[29,30] Flexible granular chains are therefore considered a model system for polymeric chains just as colloids are thought of as giant atoms.[31–36] While most studies focus on the column collapse and packing of these chains, a quasi-static measurement of the repose angle is missing despite its significance.[28,35] Here, we investigate the mechanism of repose of granular chains through a series of experiments. We delineate the influence of the laboratory protocol, geometric boundary conditions and chain length which determine the repose angle of these chains.

2 Methods

We use granular chains made of metallic (brass) spheres of diameter d = 2 mm connected with metallic links of length ~1 mm (Fig. 1a). The "I-shaped" links ensure that the surface-to-surface separation of adjacent spheres can vary between 0 and 1 mm. The chain length N is defined to be equal to the number of spheres in the chain. A simple measure of the flexibility of these chains is the diameter of the smallest loop

[a] Department of Physics, Indian Institute of Science, Bangalore, Karnataka 560012, India. E-mail: prerna@iisc.ac.in
[b] Department of Civil Engineering, Indian Institute of Science, Bangalore, Karnataka 560012, India
[c] Centre for Biosystems Science and Engineering, Indian Institute of Science, Bangalore, Karnataka 560012, India. E-mail: prerna@iisc.ac.in

† Electronic supplementary information (ESI) available. See DOI: 10.1039/d1sm01539g

This journal is © The Royal Society of Chemistry 2022

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

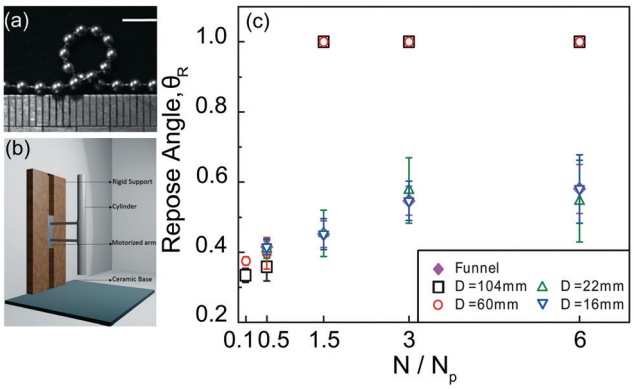

Fig. 1 (a) Image of the smallest loop formed by granular chains used in the experiment. This characteristic loop size is independent of chain length for $N > 4$. Scale bar, 5 mm. (b) Schematics of the experimental set-up to measure the repose angle by quasi-static vertical extraction of a cylinder pre-filled with chains. (c) Repose angle normalized by 90° as a function of normalized chain length obtained *via* depositing chains through a funnel and vertical extraction of a pre-filled cylinder with diameter D. The repose angle increases with an increase in chain length and cylinder diameter. The net amount/mass of granular chains used for each of the data points in this plot is kept fixed at 1.2 kg.

that can be formed by closing the chain on itself in the form of a ring. This minimum loop diameter for our experiments is ~8 mm and is independent of chain length for chains with $N > 4$. We infer the persistence length of the chains by analysing the angles between adjacent links within this smallest possible loop and find it to be $\sim 12d$ (ESI†). The number of spheres contained within a single persistence length in our experiments is $N_p \approx 8$. The maximum packing fraction of chains decreases with an increase in chain length (ESI†).

We measure the repose angle using two complementary methods: (1) deposition through a funnel while maintaining a constant drop height of ~ 2 cm from the mouth of the funnel to the top of the pile, where the width of the funnel mouth is 15 mm; (2) vertical extraction of a cylinder pre-filled with chains (Fig. 1b). We use cylinders with diameters of 16 mm, 22 mm, 60 mm, and 104 mm in the experiments, *i.e.*, the diameters are 0.7–4 times the persistence length. We also vary the total mass of chains to study the effect of the initial aspect ratio of the columns present inside the cylinders, before pulling is initiated, on the repose behaviour. The extraction speed of the cylinders in all the experiments is kept constant at ~ 1 mm s^{-1}. The time required to release the granular mass, $T_r = H/V_{lift}$, in all the experiments is much larger than the time needed to set the granular mass in motion $T_m = \sqrt{2H/g}$, where H is the filling height, V_{lift} is the pulling speed and g is the acceleration due to gravity ($T_r/T_m \sim$ 350–2500). This implies that repose is established quasi-statically.

The initial state is established by filling the cylinder with a given mass of chains, followed by manual vertical tapping. The tapping is continued until the packing fraction reaches within 25% of the maximum packing fraction that the longest chain attains for that cylinder diameter. In other words, we keep the packing fraction for all chains constant for a given cylinder diameter and total mass of chains. This allows us to generate repeatable data sets in all our experiments. For the experiments carried out using a funnel, no specific method to establish the initial state is required because a few chains are dropped through the funnel at a time. The chains are stored in a ziplock bag after the completion of experiments to keep them clean and dry for each trial.

3 Results

3.1 Deposition of granular chains through various geometries of the enclosing container

We first explore the effect of chain length on the repose angle exhibited by the chain ensembles keeping the net amount/mass of granular chains fixed. The repose angle normalized by 90°, θ_R, increases with an increase in N when measured by dropping a few chains at a time through a funnel while maintaining a constant drop height (Fig. 1c, diamonds). In these experiments, we observe the formation of a conical pile for all N.

The repose angle measured by vertically extracting a cylinder pre-filled with chains also shows an increase with an increase in chain length N (Fig. 1c). Surprisingly, in addition to dependence on N, the repose angle also depends on the diameter of the cylinder, D. Wider cylinders result in formation of stable columns of granular chains ($\theta_R \sim 1$), whereas narrower cylinders continue to form the conventional conical piles of chains ($\theta_R < 1$). The pathway of conical pile formation is fundamentally different for the two protocols used in our experiments. When using the funnel, chains fall in a geometry akin to that of a hopper and therefore naturally lead to a conical shape in the early stages of growth of the pile. As the pile grows, *i.e.* as the chains continue to drop through the funnel, the slope of the pile reaches an angle more than its steady state θ_R, at which point excess chains slump to the bottom of the pile and act as a base for new chains being poured.

In contrast, when a pre-filled cylinder of granular chains is extracted, the chains eject out of the cylinder as a plug—with the entire cross section moving at the same speed leading to formation of rigid columns ($\theta_R \sim 1$). The stability of these columns depends on their aspect ratio, *i.e.* the ratio of the height of the exposed/unsupported column, H_{exp}, to the cylinder diameter. These columns do not remain stable above a critical height, H_{cr}, and buckle when H_{exp} exceeds H_{cr}. Repeated buckling events lead to formation of a conical pile for narrower cylinders (Fig. 2a and ESI,† Movie S1). Rigid columns form in the case of wider cylinders where no buckling occurs (Fig. 2b and ESI,† Movie S2). Below a critical chain length of $N \sim 4$, θ_R does not show any dependence on the cylinder diameter D.

3.2 Flow fields during deposition

A granular flow is a key phenomenon that governs the repose behaviour. Quantitative flow fields would therefore provide insights into how introducing constraints between particles in the form of links changes the strain rates experienced by the granular ensemble under a variety of boundary conditions

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

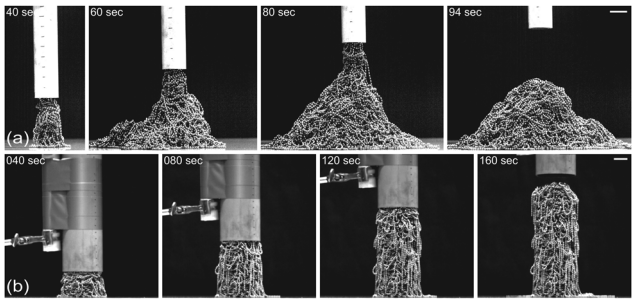

Fig. 2 Time lapse images of repose of granular chains with N = 48 filled in cylinders of diameter (a) 16 mm and (b) 60 mm. Scale bars, 2 cm. A wider cylinder results in a stable vertical column, whereas a narrower cylinder results in a conical pile with a repose angle of 61.31°. The total mass of the chains used in both (a) and (b) was the same and equal to 1.2 kg.

explored in this study. We quantitatively measure the flows of these granular ensembles using particle image velocimetry. Fig. 3b and ESI,† Movie S3 show that there is nearly no flow on the surface of the column throughout the deposition process in the case of the widest cylinder depositing long chains (N = 48). However, deposition of the same chains using narrower cylinders involved intermittent and large flows (Fig. 3a). The surface of the growing column has no flow associated with it until it begins to collapse where the flow field becomes significantly unsteady, and the column collapses into a pile (ESI,† Movie S4). The deposition continues with again a new column growing on top of this collapsed pile. Eventually this column also collapses (which we refer to as buckling instability) and the process repeats itself.

The flows associated with piles of shorter N = 4 chains are independent of the cylinder diameter and qualitatively similar to those obtained in the case of spherical particles (Fig. 4 and ESI,† Movie S5). There is a uniform downward flow all over the surface of these piles along with the magnitude of the flow velocity being largest near the top of the piles where a new material is being deposited and smallest near the base of the piles.

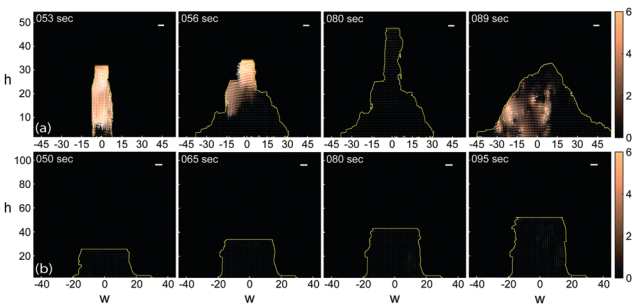

Fig. 3 Flow field at representative time points during pile deposition of chains with N = 48 through cylinders with diameters of (a) 22 mm and (b) 60 mm. Scale bars, 1 cm. Both depositions used the same total fixed mass of chains of 1.2 kg. The pile height, h, and pile width, w, are normalized with sphere diameter and the speed is normalized with the extraction speed of the cylinder. The boundaries of the piles are marked as solid yellow lines.

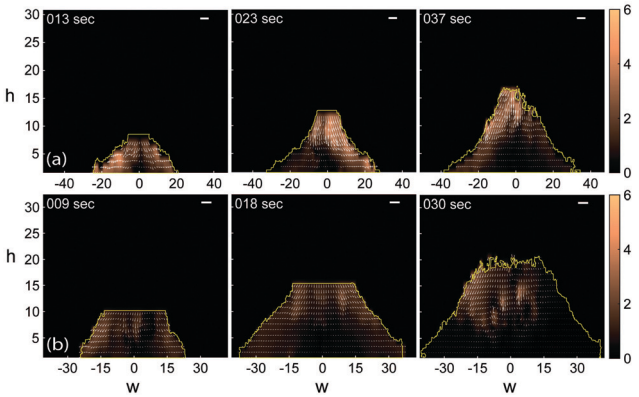

Fig. 4 Flow field at representative time points during pile deposition of chains with N = 4 through cylinders with diameters of (a) 22 mm and (b) 60 mm. Scale bar, 1 cm. Both depositions used the same total fixed mass of chains of 1.2 kg. The pile height, h, and pile width, w, are normalized with sphere diameter and the speed is normalized with the extraction speed of the cylinder. Boundaries of the piles are marked as solid yellow lines.

Fig. 5b shows that the downward component of surface flow velocity, V_y, varies linearly along the central axis of the pile during the entire period of pile formation for short N = 4 chains. That is, one can associate the slope of this curve as an "instantaneous strain rate". However, this feature breaks down when the pile is formed through repetitive buckling, for example, when long chains with N = 48 are deposited by flow through a narrow cylinder of D = 22 mm (Fig. 5a). Herein, either there are large non-monotonic surface flow velocities that are observed when the buckling instability takes place or the flow velocities are too small to be measured reliably. The instantaneous strain rates in these cases, obtained by making a linear fit close to the base of the piles, fluctuates between near zero and large values during the course of pile formation. Finally, the surface flow velocities in the case of stable column

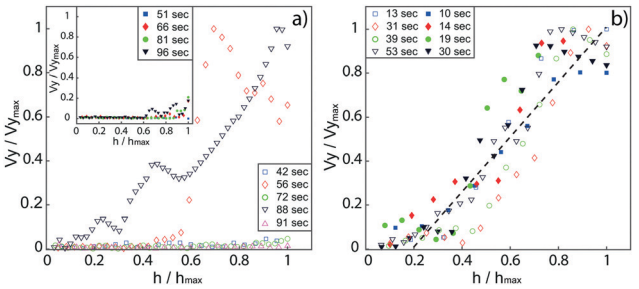

Fig. 5 Variation of the normalized vertically downward component of velocity, V_y/V_{ymax}, along the central axis of the pile at different time points during growth for (a) N = 48 and D = 22 mm; (b) N = 4 and D = 22 mm (hollow symbols) and N = 4 and D = 60 mm (solid symbols). The inset of (a) shows V_y/V_{ymax} along the central axis of the pile at different time points during growth for N = 48 and D = 60 mm wherein a stable column forms. The dashed line in (b) indicates the linear fit to the data. This linear dependence breaks down in (a) where pile formation occurs through repetitive buckling. The normalization parameters V_{ymax} and h_{max} are the highest velocity and pile height, for the particular instant for which the plot is drawn, respectively.

 This journal is © The Royal Society of Chemistry 2022

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

formation remain close to the measurement noise level throughout the process of deposition (Fig. 5a, inset).

Single spheres and chains with $N = 4$ always have larger time averaged strain rates than the longer chains for any given cylinder diameter (Fig. 6a–c). As the cylinder diameter increases, the strain rates for single spheres and chains with $N = 4$ decrease. Despite that, the observed repose angle remains the same for these systems because the flow of the ensemble is akin to a shear flow. However, for longer chains the decrease in the strain rate with an increase in cylinder diameter also coincides with a striking change in the nature of the flow itself leading to a dramatic change in the repose angle. The flow changes from being spatiotemporally unsteady and heterogenous to a uniform plug on increasing the cylinder diameter. Concomitantly, the repose angle of this final "pile" increases to $\sim 90^\circ$.

3.3 Initial filling ratio governs the pile to column transition of granular chains

The dependence of repose angle on cylinder diameter can be mapped to a more generalised parameter of the initial aspect ratio (a_i) which is equal to the ratio of the filling height (H) of the pre-filled column that resides within the cylinder, once all the chains are poured in it, to the cylinder diameter (D). Various a_i values are achieved by varying the amount/mass of granular chains that are filled into the cylinder. We find that, for chain length N greater than 4, the variation of repose angle with a_i is characteristically S-shaped (Fig. 7). That is, above a critical aspect ratio, a_{cr}, the normalized repose angle is nearly constant with a value much smaller than one and below which it saturates to exactly one. a_{cr} increases slightly with an increase in N and has a range of values between 3 and 5. These measurements reveal that the same cylinder diameter can result in either a stable column or a conical pile, for a given chain length, depending on the initial aspect ratio (ESI,† Movie S6). A chain length of $N = 4$ does not entangle sufficiently and always results in conical piles with a repose angle that is independent of a_i.

The transition from column to pile begins with buckling instability which grows into an eventual column collapse event when the height of the exposed column exceeds the critical height (H_{cr}). We therefore compare the critical aspect ratio a_{cr} found above with the ratio H_{cr}/D. We measure experimentally H_{cr} by analysing the width of the base as a function of time. This width jumps abruptly following a column collapse and the height of the pile at the time point just preceding this jump event is identified as H_{cr} (Fig. 8c). Indeed, for medium chain lengths ($N = 12$, 24 and 48) and all cylinder diameters explored in this study, we find H_{cr}/D to be 2.76, 3.22 and 3.98, respectively, which are comparable to the observed a_{cr} values of 3, 4 and 5, respectively. The other two limiting cases of pile formation by shear flow and stable column formation show linearly increasing and constant widths, respectively (Fig. 8a, b and d).

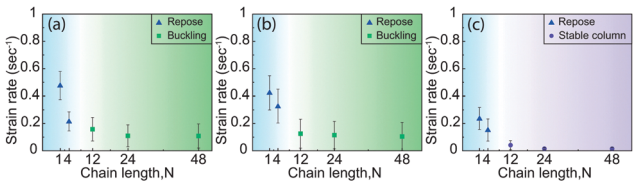

Fig. 6 Time averaged strain rate along the central vertical axis of the chain ensemble as a function of chain length for the cases in which chains with a fixed mass of 1.2 kg were extracted from cylinders of diameters (a) 16 mm, (b) 22 mm, and (c) 60 mm. Among the three growth pathways, pile formation by shear flows has the highest strain rate associated with them, while stable column formation has the lowest strain rates. Pile formation *via* repetitive buckling has intermediate strain rates. The colour and shape of data points indicate the type of flow that occur during the extraction.

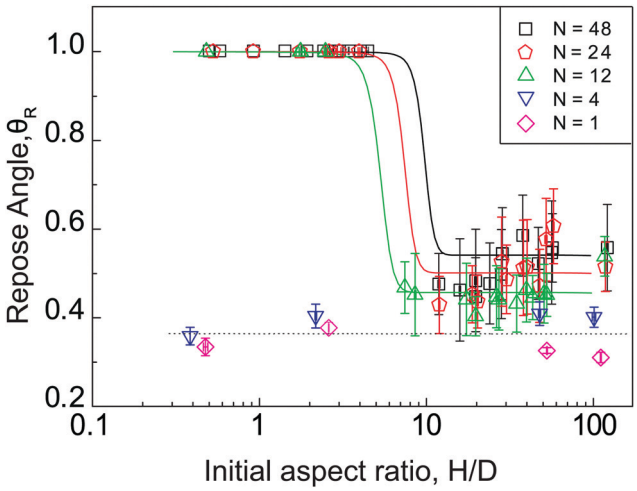

Fig. 7 Repose angle normalized by 90° as a function of the initial aspect ratio which is equal to the height up to which the cylinder is pre-filled before it is lifted up divided by the cylinder diameter. The aspect ratio is varied by changing the pre-filled mass of chains. The solid line is a guide to the eye for data corresponding to $N > 4$. The repose angle of chains with $N = 4$ and isolated spherical beads with $N = 1$ is independent of the initial aspect ratio (the dotted line is a guide to the eye for these data).

The dependence of repose on the initial aspect ratio also enables us to verify the previously developed volume conservation model which we modify to account for changes in the density of chains that occur during the process of repose.[37] The total mass of chains remains the same prior to the extraction of the cylinder and after the formation of the pile. Assuming the initial density of chains is ρ_i and the final density is ρ_f, mass conservation implies

$$\frac{R_f}{R_i} = \sqrt{\frac{6 \times a_i \times R_i}{H_f} \times \frac{\rho_i}{\rho_f}} \quad (1)$$

where R_f is the final radius of the pile that is formed after the vertical extraction of the cylinder and R_i is the radius of the column of chains that is present within the cylinder at the beginning of extraction. R_i, by definition, is equal to the radius of the cylinder. We find experimentally that the final heights of the conical piles are linearly proportional to the initial aspect ratios, $\frac{H_f}{D_i} = 10^c \times a_i^m$, with best fit values of

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

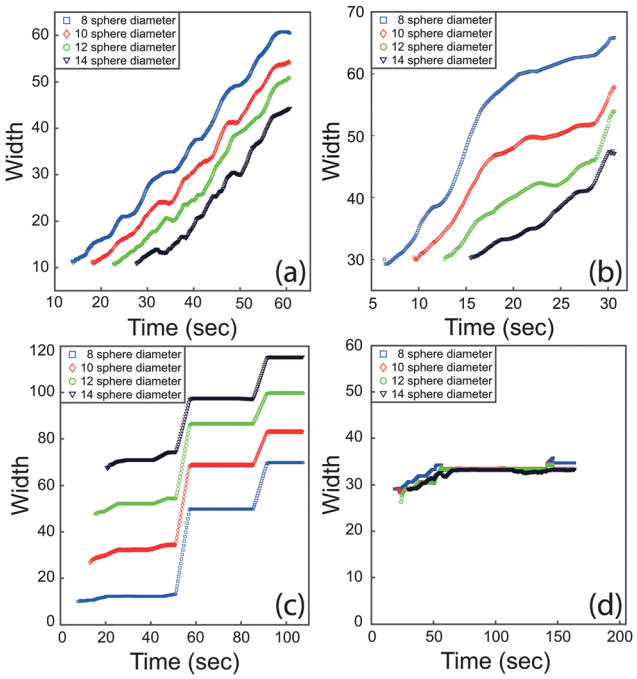

Fig. 8 Time evolution of the width of the chain ensemble at different heights from the base for (a) N = 4, D = 22 mm; (b) N = 4, D = 60 mm; (c) N = 48, D = 22 mm and (d) N = 48, D = 60 mm. (a) and (b) Correspond to conical pile formation by simple shear flow where the width of the pile increases linearly with time at early evolution times. (c) Corresponds to conical pile formation *via* repetitive buckling events and each step jump corresponds to a buckling event. Each individual plot in this panel has been offset by multiples of 20 on the Y-axis for visibility. (d) Corresponds to stable column formation, where the width stays approximately constant over time. For all panels the width has been normalized with the single sphere diameter.

$c = -0.2 \pm 0.02$ and $m = 0.4 \pm 0.02$ (Fig. 9a, solid line). Substituting this in the equation above, we get,

$$\frac{R_f}{D_i} = \sqrt{\frac{\rho_i}{\rho_f}} \times \sqrt{\frac{3}{4 \times 10^c}} \times a_i^{\frac{1-m}{2}} \tag{2}$$

This equation is then fit to the experimental data of variation of final radius of the pile with the initial aspect ratio to get best fit value of $\frac{\rho_i}{\rho_f} = 1.63 \pm 0.04$ (Fig. 9b). This implies that the dilation takes place during repose such that the final packing density ρ_f is much lower than the initial packing density ρ_i.

4 Discussion

We show that granular chains display a variety of repose behaviors depending on the boundary conditions and chain length. The transition from conical piles to stable columns takes place above a critical chain length. We conjecture that this column stability arises from the entanglement of chains. Our study also identified the critical aspect ratio of chain columns above which they would buckle and collapse into heaps or piles.

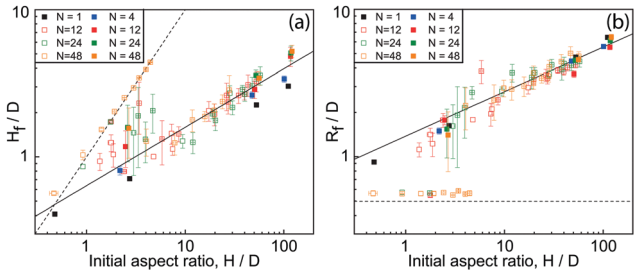

Fig. 9 (a) Height and (b) radius of the final pile normalized by cylinder diameter as a function of the initial aspect ratio. Solid symbols are data points from experiments where the mass of chains poured into the hollow cylinder was fixed to 1.2 kg irrespective of the chain length. Hollow symbols are data points from experiments where the mass of chains poured into the cylinder was varied to vary the initial aspect ratio. The dashed lines in (a) and (b) denote stable column formation wherein the final height is equal to the filling height and the final radius is equal to the radius of the cylinder, respectively. The solid line in (a) is a linear fit to the data points where conical piles formed. The solid line in (b) is a semi-empirical prediction from the mass conservation model described in Section 4.

We note that $N = 4$ chains can be approximated as rigid rods because the ratio of persistence length to contour length for these chains is larger than unity and is equal to 2. Our observation that $N = 4$ chains always form conical piles through shear flow for the various cylinder diameters and initial aspect ratios used in this study is consistent with other experiments that report stick slip behaviour for rigid rods of a similar aspect ratio.[38]

We now provide simple theoretical underpinning to our findings of the transition of repose of granular chains from columns to piles as observed during vertical extraction of pre-filled cylinders. As a preliminary examination of this process, we now use Shanley's theory of inelastic column buckling under load to predict the variation of H_{cr} with cylinder diameter.[39–41] The critical height is given by $H_{cr} = \xi\left(\frac{EI}{\rho g A}\right)^{\frac{1}{3}}$ where E is the effective elastic modulus, I is the moment of inertia of the column cross section, ρ is the mass density, g is the acceleration due to gravity, A is the area of cross section and ξ is a numerical constant. The exact value of ξ depends on the nature of load and boundary conditions. Both the load and boundary conditions remain the same for all chain lengths and cylinder diameters as long as the total mass of chains is fixed. Therefore, the ratio of critical heights corresponding to a given chain length for different cylinder diameters can easily be computed from this expression because all unknown parameters like E and ξ factor out, leaving behind quantities that are robustly known in the experiment. Indeed, the theoretical and experimentally determined ratios of critical heights for a given chain length and total mass with varying cylinder diameters are in close agreement (ESI†).

We emphasize that while entanglement of chains ushers in the stability of the columns, the overall boundary conditions can be suitably modified to access the range of responses – in

 This journal is © The Royal Society of Chemistry 2022

Published on 10 February 2022. Downloaded by Indian Institute of Science on 2/5/2024 4:24:05 AM.

effect allowing us to control entanglement of the chains. Quantification of the flow field during this repose process also shows characteristic velocity signatures during stable column formation, buckling of columns leading to piles, and standard pile formation. We identify these specific velocity signatures that are once again dependent on the applied boundary conditions in the problem.

Granular chains are an interesting and insightful model system that allows critical control of the response by modulating appropriate boundary conditions, and could be used to gain insights into the complex behavior of cohesive granular matter as observed in landslides and avalanches. They are also an interesting complement to studies on reinforcement of granular media through elastic loops and other jammed architectural structures.[23,25,42]

While we show extensive details of the kinematics of the problem, microscopic or interparticle distribution of forces within the chains and how this leads to observed repose remain an open question.[43,44]

Author contributions

TGM and PS conceptualised the study. PSS conducted the experiments. All authors analysed and interpreted the data. All authors wrote the manuscript.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgements

TGM acknowledges the DST-SERB grant no EMR-2017-003375 for funding.

Notes and references

1 T. Borzsonyi, T. C. Halsey and R. E. Ecke, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2008, **78**, 011306.
2 D. V. Denisov, K. A. Lorincz, J. T. Uhl, K. A. Dahmen and P. Schall, *Nat. Commun.*, 2016, **7**, 10641.
3 A. Daerr, *Phys. Fluids*, 2001, **13**, 2115.
4 C.-C. Liao, *Powder Technol.*, 2018, **325**(632), 638.
5 C. Li, T. Honeyands, D. O'Dea and R. Moreno-Atanasio, *Powder Technol.*, 2017, **320**, 257–272.
6 H. A. Makse, S. Havlin, P. R. King and H. E. Stanley, *Nature*, 1997, **386**, 379–382.
7 J. Baxter, U. Tuzun, D. Heyes, I. Hayati and P. Fredlund, *Nature*, 1998, **391**, 136.
8 G. Riley and G. Mann, *Mater. Res. Bull.*, 1972, 7(163), 169.
9 B.-B. Dai, J. Yang, C.-Y. Zhou and W. Zhang, *Proceedings of the 7th International Conference on Discrete Element Methods*, Singapore, 2017, pp. 767–776.
10 A. Samadani and A. Kudrolli, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2001, **64**, 051301.
11 N. Mitarai and F. Nori, *Adv. Phys.*, 2006, **55**, 1–45.
12 D. J. Hornbaker, R. Albert, I. Albert, A.-L. Barabási and P. Schiffer, *Nature*, 1997, **387**, 765.
13 D. C. Shorts and K. Feitosa, *Granular Matter*, 2017, **20**, 2.
14 Y. Zhou, B. Xu, A. Yu and P. Zulli, *Powder Technol.*, 2002, **125**(45), 54.
15 Y. C. Zhou, B. H. Xu, A. Yu and P. Zulli, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2001, **64**, 021301.
16 B.-B. Dai, J. Yang and C.-Y. Zhou, *Granular Matter*, 2017, **19**, 24.
17 H. Wei, X. Tang, Y. Ge, M. Li, H. Saxén and Y. Yu, *Powder Technol.*, 2019, **353**, 526–534.
18 H. M. Beakawi Al-Hashemi and O. S. Baghabra Al-Amoudi, *Powder Technol.*, 2018, **330**, 397–417.
19 P. Bak, C. Tang and K. Wiesenfeld, *Phys. Rev. Lett.*, 1987, **59**, 381–384.
20 D. Dhar, *Phys. A*, 2006, **369**, 29–70.
21 G. A. Held, D. H. Solina, H. Solina, D. T. Keane, W. J. Haag, P. M. Horn and G. Grinstein, *Phys. Rev. Lett.*, 1990, **65**, 1120–1123.
22 K. J. Dong, R. P. Zou, K. W. Chu, R. Y. Yang, A. B. Yu and D. S. Hu, *AIP Conf. Proc.*, 2013, **1542**, 646–649.
23 N. Gravish, S. V. Franklin, D. L. Hu and D. I. Goldman, *Phys. Rev. Lett.*, 2012, **108**, 208001.
24 K. Murphy, N. Reiser, D. J. Choksy, C. E. Singer and H. M. Jaeger, *Granular Matter*, 2015, **18**, 1–9.
25 S. Franklin, *Phys. Today*, 2012, **65**, 70.
26 S. V. Franklin, *Europhys. Lett.*, 2014, **106**, 58004.
27 M. Trepanier and S. V. Franklin, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2010, **82**, 011308.
28 L.-N. Zou, X. Cheng, M. L. Rivers, H. M. Jaeger and S. R. Nagel, *Science*, 2009, **326**, 408–410.
29 P. G. de Gennes, *Scaling Concept in Polymer Physics*, Cornell University Press, Ithaca, New York, 1979.
30 M. Doi and S. F. Edwards, *The Theory of Polymer Dynamics*, Clarendon Press, Oxford, 1988.
31 K. Safford, Y. Kantor, M. Kardar and A. Kudrolli, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2009, **79**, 061304.
32 F. Samadi Taheri, H. Fazli, M. Doi and M. Habibi, *Soft Matter*, 2018, **14**, 5420–5427.
33 W. C. K. Poon, *J. Phys. A: Math. Theor.*, 2016, **49**, 401001.
34 A. E. Larsen and D. G. Grier, *Phys. Rev. Lett.*, 1996, **76**, 3862–3865.
35 D. Dumont, M. Houze, P. Rambach, T. Salez, S. Patinet and P. Damman, *Phys. Rev. Lett.*, 2018, **120**, 088001.
36 P. Wen, G. Wang, D. Hao, N. Zheng, L. Li and Q. Shi, *Phys. A*, 2015, **419**, 457.
37 E. Lajeunesse, A. Mangeney-Castelnau and J. P. Vilotte, *Phys. Fluids*, 2004, **16**, 2371–2381.
38 K. Desmond and S. V. Franklin, *Phys. Rev. E: Stat., Nonlinear, Soft Matter Phys.*, 2006, **73**, 031306.
39 F. R. Shanley, *J. Aeronaut. Sci.*, 1947, **5**, 261–268.
40 S. Karmakar, A. Sane, S. Bhattacharya and S. Ghosh, *Phys. Rev. E*, 2017, **95**, 042903.
41 A. G. Greenhill, *Proc. Cambridge Philos. Soc.*, 1881, **4**, 65–73.
42 A. Guerra and D. P. Holmes, *Soft Matter*, 2021, **17**, 7662–7669.
43 E. T. Owens and K. E. Daniels, *EPL*, 2011, **94**, 54005.
44 E. Lerner, G. Düring and M. Wyart, *Proc. Natl. Acad. Sci. U. S. A.*, 2012, **109**, 4798–4803.